

Prikshit Gautam

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🐙 GitHub – 🔗 LinkedIn – 🌐 Portfolio – 📝 Medium

Education

Thapar Institute of Engineering and Technology, Patiala

B.E. Electronics and Computer Engineering, CGPA: 8.46

Aug 2023 – Jun 2027

St. Soldier Divine Public School, Nangal

XII (CBSE); **92.67%**

May 2022

Technical Skills

Languages:	Python, C++, MATLAB Programming, SQL
Frameworks:	PyTorch, Scikit-learn, keras, NumPy, Pandas, Matplotlib, Tensorflow, PyMuPDF, Sentence-Transformers, Seaborn
Developer Tools:	Git, MySQL, VS Code, Streamlit, Node.js, AWS, Hugging Face, GitHub, Google Colab, VS Code, Postman, MATLAB
AI Technology:	NLP, RAG, Time Series Forecasting, Data Visualization, OpenCV (YOLOv8)
Coursework:	Data Structures and Algorithms, Operating Systems, Database Management System, Object Oriented Programming, Cloud Computing Technology, Blockchain Technologies, Computer Networks

Experience

EduNet Foundation | LLM Intern | [Credentials](#)

Jan 2025 - Feb 2025

- Spearheaded the development of a Python NLP chatbot, boosting automated **query resolution by 30%** and achieving over **90% intent accuracy**.
- Reduced average API response **latency by 25%** by optimizing the model's inference pipeline and deployment configuration.
- Technologies:** Python, LangChain, Hugging Face, Groq, FAISS (Vector DB), PyMuPDF, Streamlit.

Experiential Learning Centre, TIET | Summer Project Intern | [Credentials](#)

Jun 2025 – Jul 2025

- Developed a computer vision system detecting 2 distinct roadway hazards from real-time video using **YOLOv8**, decreasing model inference time by **15 milliseconds** while maintaining **89.7% mAP**.
- Presented project outcomes and a live demonstration to a panel of 10+ faculty members, earning commendation for system effectiveness and real-world viability.
- Technologies:** Python, YOLOv8, Computer Vision, Sightengine API.

Projects

5G Latency Predictor | Pytorch, Keras, Statistical analysis | [LINK](#) | [LIVE](#)

Aug 2024 – Jul 2025

- Analyzed 40,000+ real-world 5G packet traces from **KTH Expeca testbed (Sweden)**, identifying RLC frame alignment as root cause of latency.
- Designed and benchmarked three neural architectures (RNN, LSTM, Hybrid LSTM), achieving an **accuracy of 95%** in forecasting IP delay.
- Reduced prediction variance by 45% using a multi-step hybrid LSTM approach.

AI-RoadIntelligence | YOLOv8, OpenCV, SUMO, SightEngine API | [LINK](#)

Jun 2025 – Jul 2025

- Engineered a real-time traffic congestion reduction system using YOLOv8, processing video inputs and concurrent **YouTube live streams at 45 FPS** with less than 50ms processing latency.
- Programmed core product logic to trigger automatic **green light extensions of 10 – 15 seconds**, mitigating potential accidents post-hazard detection.
- Optimized emergency routing system by integrating ML models with SUMO, decreasing emergency **vehicle arrival times by 12%** during trials on the test grid and real-world simulations

Healthcare Assistant Chatbot | Python, LangChain, Hugging Face | [LINK](#) | [LIVE](#)

Jan 2025 – Feb 2025

- Implemented a LangChain RAG pipeline with FAISS vector store, improving **clinical query relevance by 40%**.
- Fine-tuned Hugging Face transformer models for domain-specific intent classification, enhancing **accuracy across 2,050+ medical topics**.
- Curated an interactive **web application with Streamlit** that provided vital health parameters and emergency contact data, leading to an increase in user satisfaction scores and enhancing the overall patient experience by 20%.

Achievements & Certifications

- Adobe Hackathon (Top 4029/115,000 Teams)** Unstop, Aug 2025
Constructed Smart PDF Reader with semantic linking. [Repository](#)
- Complete Data Science and Machine Learning** Issued: Jul 2025
Udemy Instructor: Krish Naik [Certificate](#)
- Geometrical Shape Detection using Python** Issued: Apr 2024
Experiential Learning Centre, TIET - Verified by Thapar-LMS [Certificate](#)